

Normalization

It was an important process in our database design. It helps us to improve the database's efficiency, consistency, and accuracy. It makes it easier to manage and maintain the data and ensures that the database is adaptable to changing business needs in the real estate business.

In order to achieve this stage we studied deeply about normalization and its types upto BCNF and apply them on our design.

1. 1NF (First Normal Form)

Rules for 1NF are easy.” All attributes are single valued!”.

In order to achieve this rule we must do two things. First identify if there is any problem. In our table phone has multiple values and email also has multiple values. We did this to make it efficient for both tenants and owners with sustainable contact information. Second fix the problem. We achieve that by splitting the multivalued attributes.

The result becomes:

USER(user_id(PK), full name, password, name)

USER_PHONE(phone_id (PK), user_id (FK) , phone)

USER_EMAIL(email_id (PK), user_id (FK), email)

Now 1NF is done successfully!!!

2. Second Normal Form (2NF)

Rule for 2NF is “Should be on 1NF and no partial dependency”.

1NF was done before, and for the second rule even on composite ID like user, user phone and user email we don't have partial dependency we have full dependency. Now we can surely say that 2NF is done successfully and proceed.

3. Third Normal Form (3NF)

The rule for 3NF is “Should be on 2NF and no transitive property”.

In our design both are satisfied.

- USER → depend on user_id
- PROPERTY → depends on property_id
- LOCATION → separated from user

- USER_PHONE → phone depends only on phone_id

4. BCNF (Boyce-Codd Normal Form)

Rule for BCNF is “Satisfy 3NF and Every determinant must be a candidate key”.

In our system, the design satisfies BCNF as all determinants are candidate keys and there are no functional dependencies violating BCNF conditions.

NORMALIZATION SUMMARY TABLE

Normal Form	Rule	Issue in Initial Design (A)	Solution Applied	Result
1NF (First Normal Form)	All attributes must be atomic. No repeating groups or multivalued attributes.	• phone stored as multiple values (e.g., "0911, 0922") in one field. • email stored as multiple values (e.g., "a@mail.com, b@mail.com") in one field.	Decomposed multivalued attributes into separate tables: • USER_PHONE (phone_id, user_id, phone) • USER_EMAIL (email_id, user_id, email)	All attributes are atomic. Each table contains single values only. Satisfies 1NF.
2NF (Second Normal Form)	Must be in 1NF. No partial dependency of any non-key attribute on a part of a composite key.	No issue. All tables have single attribute primary keys (user_id, phone_id, email_id, property_id, location_id, etc.).	No change needed.	No partial dependencies exist. Satisfies 2NF.
3NF (Third Normal Form)	Must be in 2NF. No transitive dependency (non-key attribute should not depend on another non-key attribute).	In PROPERTY table (if location was not separated): • city → subcity • property_id → city → subcity This is a transitive dependency.	Created separate LOCATION table: LOCATION (location_id, city, subcity) PROPERTY now stores only location_id (FK).	Transitive dependency removed. All non-key attributes depend only on the primary key. Satisfies 3NF.
BCNF (Boyce-Codd Normal Form)	For every functional dependency $X \rightarrow Y$, X must be a candidate key (super key).	No determinant violations observed in the decomposed tables.	All tables designed so that every determinant is a candidate key.	All determinants are candidate keys. Satisfies BCNF.
Conclusion The initial design contained multivalued attributes and potential transitive dependencies. Through normalization, the schema was decomposed into multiple relations to eliminate redundancy and ensure data integrity. The final design satisfies all normalization requirements up to BCNF.				

In general now shout out to our design, it is normalized to BCNF. Multivalued attributes such as phone numbers and email addresses have been separated into independent relations to satisfy (1NF). All relations have single primary keys, eliminating partial dependencies (2NF), and transitive dependencies have been removed (3NF). Furthermore, all determinants are candidate keys, ensuring compliance with BCNF.